Chapter 7 Homework

Part A Multiple choice--Select the one that is best in each case.

1. Electrons in the 1*s* subshell are much closer to the nucleus in Ar than in He due to the larger ________ in Ar.
A) nuclear charge
B) paramagnetism
C) diamagnetism
D) Hund's rule
E) azimuthal quantum number

2. The __________ have the most negative electron affinities.
A) alkaline earth metals
B) alkali metals
C) halogens
D) transition metals
E) chalcogens

3. Consider the following electron configurations to answer the questions that follow:
(i) $1s^2\ 2s^2\ 2p^6\ 3s^1$
(ii) $1s^2\ 2s^2\ 2p^6\ 3s^2$
(iii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
(iv) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
(v) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
The electron configuration belonging to the atom with the highest <u>second</u> ionization energy is ________.
A) (i) B) (ii) C) (iii) D) (iv) E) (v)

4. The ground-state electron configuration of a Cr atom is $[Ar]4s^13d^5$. What is the electron configuration of a Cr^{3+} ion?
A) $[Ar]4s^13d^2$ B) $[Ar]4s^13d^8$ C) $[Ar]4s^23d^1$ D) $[Ar]3d^3$ E) $[Ar]3d^9$

5. ________ has been shown to form compounds only when it is combined with something with an extremely high ability to remove electrons from other substances.
A) Neon B) Helium C) Hydrogen D) Xenon E) Oxygen

6. In general, as you go across a period in the periodic table from left to right:

(i) the atomic radius ________;
(ii) the effective nuclear charge ________; and
(iii) the first ionization energy ________.

A) decreases, decreases, increases
B) increases, increases, decreases
C) increases, increases, increases
D) decreases, increases, decreases
E) decreases, increases, increases

7. Element M reacts with oxygen to form an oxide with the formula MO. When MO is dissolved in water, the resulting solution is basic. Element M could be ________.
A) Cesium
B) Strontium
C) Rubidium
D) Germanium
E) Nitrogen

8. In which orbital does an electron in a nitrogen atom experience the greatest shielding?
A) 3*p* B) 3*s* C) 2*p* D) 2*s* E) 1*s*

9. Which alkaline earth metal will not react with liquid water or with steam?
A) Be B) Mg C) Ca D) Ba
E) They all react with liquid water and with steam.

10. Sodium is much more likely to exist as a cation than is chlorine. This is because ____.
A) chlorine is a gas and sodium is a solid
B) chlorine has a greater electron affinity than sodium does
C) chlorine is bigger than sodium
D) chlorine has a greater ionization energy than sodium does
E) chlorine is more metallic than sodium

11. Based on the ionization energies of element X given in the table below, which of the following is most likely the empirical formula of an oxide of element X?

	Ionization Energy (kJ/mol)
First	577
Second	1816
Third	2745
Fourth	11577
Fifth	14482

A) XO_2 B) X_2O C) X_2O_3 D) X_2O_5 E) X_3O_2

12. Of the following metals, ________ exhibits multiple oxidation states.
A) Ba B) Rb C) Sr D) Cs E) Cr

13. When two elements combine to form a compound, the ________ the difference in metallic character between the two elements, the ________ the likelihood that the compound will be a ________ at room temperature.
A) smaller, greater, solid
B) greater, greater, liquid
C) greater, greater, solid
D) smaller, greater, liquid
E) greater, smaller, solid

14. ________ is isoelectronic with argon, and ________ is isoelectronic with neon.
A) P^{2-}, N^{2-} B) P^{3-}, N^{3-} C) P^{3+}, N^{3+} D) N^{3-}, P^{3-} E) P, N

15. Which of the following is not true about oxygen?
A) The most stable allotrope is O_3.
B) Oxygen forms two unstable ions, peroxide and superoxide.
C) About 21% of dry air consists of O_2 molecules.
D) Ozone is toxic, whereas O_2 is not.
E) Ozone absorbs certain wavelengths of ultraviolet light that O_2 does not.

Part B Short questions (20 points)

16. (3 points) **(a)** Based on the ionization energies of the alkali metals, which alkali metal would you expect to undergo the most exothermic reaction with chlorine gas? **(b)** Write a balanced chemical equation for the reaction in part **(a)**.

17. (4 points) Consider S, Cl, and K and their most common ions.
(a) List the atoms in order of increasing size.
(b) List the ions in order of increasing size.

18. (4 points) **(a)** If the core electrons completely screen the valence electrons and the valence electrons provided no screening for each other, what would be the effective nuclear charge acting on a valence electron in sulfur? **(b)** Repeat these calculations using Slater's rules. Express your answer using two decimal places.

19. (5 points) Consider the first ionization energy of the fluoride ion, F^- and the electron affinity of the neutral atom, F.
(a) Write equations for each process.
(b) These two quantities have opposite signs. Which will be positive, and which will be negative?
(c) Would you expect the magnitudes of these two quantities to be equal? If not, which one would you expect to be larger?

20. (4 points) **(a)** Why is calcium generally more reactive than beryllium? **(b)** Why does carbon atom has more negative electron affinity than nitrogen atom?

Chapter 7 Homework Answer Sheet

Name: **Student ID:** **Instructor:** **Score:**

Part A Multiple choice (15 points)

1-5______________

6-10_____________

11-15____________

Part B Short questions (20 points)